Construct List using XOR Queries

Difficulty: **Medium** Accuracy: **50.86%** Submissions: **45K+**
Points: **4** Average Time: **20m**

There is an array that initially contains only a single value, **0**.

Given a list of queries **queries[][]** of size **q**, where each query is of one of the following types:

- **0 x**: Insert x into the array.
- **1 x**: Replace every element **a** in the array with **a ^ x**, where ^ denotes the bitwise XOR operator.

Return the array in sorted order after performing all the queries.

Examples:

Input: q = 5, queries[] = [[0, 6], [0, 3], [0, 2], [1, 4], [1, 5]]
Output: [1, 2, 3, 7]
Explanation:
[0] (initial value)
[0, 6] (add 6 to list)
[0, 6, 3] (add 3 to list)
[0, 6, 3, 2] (add 2 to list)
[4, 2, 7, 6] (XOR each element by 4)
[1, 7, 2, 3] (XOR each element by 5)

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```python
class Solution:
    def constructList(self, queries):
        xr = 0
        arr = [0]

        for typ, x in queries:
            if typ == 0:
                arr.append(x ^ xr)
            else:
                xr ^= x

        ans = [x ^ xr for x in arr]
        ans.sort()
        return ans
```

The sorted list after performing all the queries is [1, 2, 3, 7].

Custom Input Compile & Run Submit